

Small-time Controllability of Control Systems on 3D Lie Groups

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Abstract

We study affine control systems with one control on 3D unimodular Lie groups. Such control systems are first classified into a finite number of equivalent systems. After completing this classification, we determine the small-time controllability of each of the equivalent systems.

1 Introduction

In [1], Agrachev, Kazandjian, and Pozzoli considered the Schrödinger equation for the quantum harmonic oscillator, given by

$$\begin{cases} i\partial_t\psi(t, x) &= \left(-\frac{1}{2}\Delta + u_0(t)\frac{|x|^2}{2} + \sum_{j=1}^d u_j(t)x_j\right)\psi(t, x) \\ \psi(t=0) &= \psi_0 \in L^2(\mathbb{R}^d, \mathbb{C}) \end{cases} \quad (1)$$

where the control $u = (u_0, \dots, u_d) \in PWC([0, T], \mathbb{R}^{d+1})$. This equation is highly relevant in quantum dynamics, with applications in physics and chemistry. Based on this equation, the paper studied the controllability of left-invariant control affine systems of the form

$$\frac{d}{dt}q(t) = q(t)\left(a + u_0(t)b + \sum_{i=1}^d u_i(t)X_i + r(t)Z\right) \quad (2)$$

where $q \in SL_2 \ltimes H_d$. Notably, the paper proved that equation (2) is small-time controllable on $SL_2 \ltimes H_d$, and that equation (2) with $u_i, r = 0$ for $i \in \{1, \dots, d\}$ is small-time controllable on SL_2 .

This result motivates the question: does there exist other 3D unimodular Lie groups for which the statement is true? To answer this question, we study affine control systems with one control on 3D unimodular Lie groups, taking the form

$$\dot{q}(t) = (A + u(t)B)q(t) \quad (3)$$

with $q \in G$, $A, B \in \mathfrak{g}$, and u the control. G is a 3D unimodular Lie group, which is known to be completely classified into the Lie groups $SU(2)$, $SL(2)$, H , $SE(2)$, and $SH(2)$. For each, $\mathfrak{g}$ is the corresponding Lie algebra. In section 2, we first classify such systems on each of these Lie groups into equivalent systems. Apparently, there are only finitely many such systems, 22 to be precise. After this classification is done, we will study controllability. We will begin section 3 by formally defining small-time controllability, among other concepts from geometric control theory. Then, we will characterise the small-time controllability of each of the equivalent systems found. This is a thorough process that will give an answer to the question posed. We see that there is only one model system that is small-time controllable.

The following table summarises the key results of this paper, and we will refer back to it later.

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Lie group	Model systems	STC
$SU(2)$	(σ_1, σ_3)	No
$SL(2)$	(E, H)	No
	(F, H)	No
	$(E + F, H)$	No
	$(E, E - F)$	No
	(H, E)	No
	(F, E)	Yes
H	(X, Z)	No
	(Z, X)	No
	(Y, X)	No
$SE(2)$	(Y, X)	No
	(Z, X)	No
	(X, Z)	No
$SH(2)$	(X, Z)	No
	(Y, Z)	No
	$(X + Y, Z)$	No
	$(X, X + Y)$	No
	$(Z, X + Y)$	No
	(Z, Y)	No
	(X, Y)	No
	(Z, X)	No
	(Y, X)	No

Table 1: Summary of Model Systems and Small-Time Controllability

1.1 Convention

Recall the following definitions of the 3D unimodular Lie groups that will be used throughout the paper.

- $SU(2) = \left\{ \begin{pmatrix} a & -\bar{b} \\ b & \bar{a} \end{pmatrix} : a, b \in \mathbb{C}, |a|^2 + |b|^2 = 1 \right\}$
- $SL(2) = \{X \in M_2(\mathbb{R}) : \det X = 1\}$
- $H = \left\{ \begin{pmatrix} 1 & a & c \\ 0 & 1 & b \\ 0 & 0 & 1 \end{pmatrix} : a, b, c \in \mathbb{R} \right\}$
- $SE(2) = \left\{ \begin{pmatrix} \cos \theta & -\sin \theta & x \\ \sin \theta & \cos \theta & y \\ 0 & 0 & 1 \end{pmatrix} : x, y, \theta \in \mathbb{R} \right\}$
- $SH(2) = \left\{ \begin{pmatrix} \cosh \theta & \sinh \theta & x \\ \sinh \theta & \cosh \theta & y \\ 0 & 0 & 1 \end{pmatrix} : x, y, \theta \in \mathbb{R} \right\}$

The following are the corresponding Lie algebras and the basis elements chosen for each.

- $\mathfrak{su}(2)$: $\sigma_1 = \frac{1}{2} \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix}$, $\sigma_2 = \frac{1}{2} \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$, $\sigma_3 = \frac{1}{2} \begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix}$
- $\mathfrak{sl}(2)$: $E = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$, $F = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$, $H = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$

- $\mathfrak{h}$: $X = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$, $Y = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}$, $Z = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$
- $\mathfrak{se}(2)$: $X = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$, $Y = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}$, $Z = \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$
- $\mathfrak{sh}(2)$: $X = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$, $Y = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}$, $Z = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$

2 Classifying Control Systems

Consider the control system (3) with $q \in G$ and $A, B \in \mathfrak{g}$, as formulated in the introduction. When classifying (3) on a particular Lie group, we want to simplify the pair (A, B) into an equivalent pair, say (f_0, f_1) . This equivalency is denoted $(A, B) \sim (f_0, f_1)$.

What tools are available for this classification? The first are inner automorphisms on A and B , the mapping $\Phi : \mathfrak{g} \rightarrow \mathfrak{g}$ which acts by conjugation by some $g \in G$. The Killing form, denoted $B(\cdot, \cdot)$, is invariant,

$$B(gAg^{-1}, gBg^{-1}) = B(A, B)$$

so $(A, B) \sim (\Phi(A), \Phi(B))$. More generally, this equivalence relation is also true for outer automorphisms $\Phi : \mathfrak{g} \rightarrow \mathfrak{g}$. Second, the control can be shifted by δ , so

$$A + uB = A + (u' + \delta)B = (A + \delta B) + u'B$$

The control can further be scaled by a constant μ , giving

$$A + uB = A + (\mu u')B = A + u'(\mu B)$$

Finally, the time can be scaled by a factor of $1/\lambda$ so that $(A, B) \sim (\lambda A, \lambda B)$. The equivalence relations are summarised in the following lemma.

Lemma 2.1. For the control system (3), we have the following equivalence relations.

1. Inner, outer automorphisms: $(A, B) \sim (\Phi(A), \Phi(B))$
2. Control shifting: $(A, B) \sim ((A + \delta B), B)$
3. Control scaling: $(A, B) \sim (A, \mu B)$
4. Time scaling: $(A, B) \sim (\lambda A, \lambda B)$

Similar work regarding the classification of control systems may have already been partially done by C. Remsing, for example in [3], [4], [5]. They notabaly worked on the $SU(2)$, $SL(2)$, H , and $SE(2)$ groups, but used Lie group isomorphisms and Lie algebra isomorphisms, unlike what we are doing in this paper. This may cause the results to differ.

Remark. We will always assume that B is nonzero, and that A and B are independent.

Proposition 2.2. Control systems of the form (3) on each 3D unimodular Lie group are equivalent to one of the model systems for the 3D unimodular Lie group, as listed in Table 1.

Proof. The demonstration for this proposition is given throughout the following sections. $\square$

2.1 Classifying Control Systems on $SU(2)$

We consider the system (3) with $q \in SU(2)$ and $A, B \in \mathfrak{su}(2)$. Begin by writing A and B as linear combinations of the basis elements

$$A = a_1^0 \sigma_1 + a_2^0 \sigma_2 + a_3^0 \sigma_3, \quad B = b_1^0 \sigma_1 + b_2^0 \sigma_2 + b_3^0 \sigma_3.$$

In this case, $B(A, A)$ and $B(B, B)$ represent the lengths of A and B respectively, and $B(A, B)$ represents the angle between A and B (all in the sense of the usual Euclidean geometry). One can check that these quantities are preserved under conjugation, suggesting that inner automorphisms act as rotations in $\mathbb{R}^3$. Alternatively, we can see this by noting

$$\text{Aut}(\mathfrak{su}(2)) = SO(3)$$

So, we construct a 3D space with $\sigma_1, \sigma_2, \sigma_3$ as the axes, analogous to $\mathbb{R}^3$. Applying a combination of conjugation group actions to both A and B , we can rotate B onto the σ_3 axis, then rotate A onto the σ_1 - σ_3 plane.

$$\begin{aligned} (A, B) &\sim (gAg^{-1}, gBg^{-1}) \\ &= (a_1^1 \sigma_1 + a_3^1 \sigma_3, b_3^1 \sigma_3) \\ &\sim (a_1^1 \sigma_1, b_3^1 \sigma_3) \\ &\sim (a_1^1 \sigma_1, a_1^1 \sigma_3) \\ &\sim (\sigma_1, \sigma_3) \end{aligned}$$

Hence, all control systems (3) on $SU(2)$ are equivalent to (σ_1, σ_3) .

2.2 Classifying Control Systems on $SL(2)$

We consider the system (3) with $g \in SL(2, \mathbb{R})$ and $A, B \in \mathfrak{sl}(2)$. Recall that for matrices $u_i, u_j \in \mathfrak{sl}(2)$, the matrix of the Killing form is the matrix K whose entries are $k_{ij} = B(u_i, u_j)$. Using the aforementioned basis elements, we compute the Killing form matrix to be

$$K = \begin{pmatrix} 0 & 4 & 0 \\ 4 & 0 & 0 \\ 0 & 0 & 8 \end{pmatrix}$$

The eigenvalues of this matrix are -4, 4, and 8. There are 2 positive eigenvalues and 1 negative eigenvalue, implying a metric signature (2, 1). So $SL(2)$ is a group of transformations in Lorentzian space $\mathbb{R}^{2,1}$. The Killing form is analogous to the Minkowski scalar product.

In Lorentzian space, vectors can be classified as elliptic, hyperbolic, or parabolic based on the sign of their norm. Let $X \in \mathfrak{sl}(2)$. If $B(X, X) = 0$, then X is parabolic, lying on the light cone. If $B(X, X) < 0$, then X is elliptic, lying on the hyperboloid of two sheets inside the light cone. If $B(X, X) > 0$, then X is hyperbolic, lying on the hyperboloid of one sheet outside the light cone.

Lorentz transformations - including null rotations, rotations, and Lorentz boosts - act via inner automorphisms. There does not exist a Lorentz transformation between the hyperboloid of two sheets and the hyperboloid of one sheet. There also does not exist a Lorentz transformation between the upper and lower sheets of the hyperboloid of two sheets.

The strategy is to identify a representative for each class. We choose $E, H, E - F$ as the representatives for the parabolic, hyperbolic, and elliptic cases, respectively. B could be in any of these classes, giving rise to three cases to be analysed. Since this representative and B lie in the same region of space, we can apply Lorentz transformations via conjugation to simplify B to this representative. We may then proceed to simplify A .

Remark. If, in our convention, E is the representative for the upper half of the light cone, then $-E$ is the representative for the lower half of the light cone. The same goes for the hyperboloid of two sheets, by taking $E - F$ and $F - E$. However, it is not necessary to discuss the upper and lower halves separately. One can be transformed to the other by scaling the control.

Begin by writing A and B as linear combinations of the basis elements

$$A = a_1^0 E + a_2^0 F + a_3^0 H, \quad B = b_1^0 E + b_2^0 F + b_3^0 H.$$

Case 1: $B(B, B) > 0$, B is hyperbolic

By a Lorentz transformation, B can be simplified to a multiple of H .

$$\begin{aligned} (A, B) &\sim (gAg^{-1}, gBg^{-1}) \\ &= (a_1^1 E + a_2^1 F + a_3^1 H, b_3^1 H) \end{aligned}$$

If $a_1^1 \neq 0$ and $a_2^1 = 0$, then

$$\begin{aligned} (A, B) &\sim (a_1^1 E + a_2^1 F + a_3^1 H, b_3^1 H) \\ &= (a_1^1 E + a_3^1 H, b_3^1 H) \\ &\sim (a_1^1 E, b_3^1 H) \\ &\sim (a_1^1 E, a_1^1 H) \\ &\sim (E, H) \end{aligned}$$

If $a_1^1 = 0$ and $a_2^1 \neq 0$, then

$$\begin{aligned} (A, B) &\sim (a_1^1 E + a_2^1 F + a_3^1 H, b_3^1 H) \\ &= (a_2^1 F + a_3^1 H, b_3^1 H) \\ &\sim (a_2^1 F, b_3^1 H) \\ &\sim (a_2^1 F, a_2^1 H) \\ &\sim (F, H) \end{aligned}$$

If $a_1^1 \neq 0$ and $a_2^1 \neq 0$, then we need to find elements in the stabiliser group of H to continue simplifying while preserving H . Such elements of $Stab(H)$ must be of the form

$$g = \begin{pmatrix} a & 0 \\ 0 & d \end{pmatrix}, \quad ad = 1$$

which represents a Lorentz boost. Applying to $(a_1^1 E + a_2^1 F + a_3^1 H, b_3^1 H)$, we have

$$\begin{aligned} (A, B) &\sim (a_1^1 E + a_2^1 F + a_3^1 H, b_3^1 H) \\ &\sim (a_1^1 a^2 E + a_2^1 d^2 F + a_3^1 H, b_3^1 H) \\ &\sim (c(E + F), b_3^1 H) \\ &\sim (c(E + F), cH) \\ &\sim (E + F, H) \end{aligned}$$

where we chose a and d such that $a_1^1 a^2 = a_2^1 d^2 = c$. The precise choices for a and d can be found by using $a_1^1 a^2 = a_2^1 d^2$ along with $ad = 1$.

Taking note of the sign, there is also $(-E - F, H)$, but this can be transformed into $(E + F, H)$ be an outer automorphism. Precisely, we have

$$Aut(\mathfrak{sl}(2)) \cong PGL(2, \mathbb{R})$$

so we can take

$$D = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

and perform the mapping $g \mapsto DgD^{-1}$ for $g \in \mathfrak{sl}(2)$.

Case 2: $B(B, B) < 0$, B is elliptic

By a Lorentz transformation, B can be simplified to a multiple of $E - F$.

$$\begin{aligned} (A, B) &\sim (gAg^{-1}, gBg^{-1}) \\ &= (a_1^1 E + a_2^1 F + a_3^1 H, b_1^1 (E - F)) \end{aligned}$$

To continue, we need elements of the stabiliser group $Stab(E - F)$. These elements take the form

$$g = \begin{pmatrix} a & b \\ -b & a \end{pmatrix}, \quad a^2 + b^2 = 1$$

which resembles a rotation matrix, and so the Lorentz transformations in this case act as rotations in a plane. If we fix $E - F$ as one axis, then for a 3D space, we need to construct two more, and it is on the plane formed by these two axes that we will rotate. Consider

$$E + F = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad H = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$$

By computing the Killing forms, one can check that $E - F$, $E + F$, and H are indeed orthogonal, forming a set of coordinate axes. Finding these two remaining axes can be thought of as finding the orthogonal complement of B , $B^\perp$, spanned by $E + F$ and H . Now we rewrite A and B in terms of this new set of coordinates, reduce A onto $B^\perp$ by control shifting, then rotate on $B^\perp$. That is,

$$\begin{aligned} (A, B) &\sim (a_1^1 E + a_2^1 F + a_3^1 H, b_1^1 (E - F)) \\ &\sim (a_1^2 (E + F) + a_3^2 H, b_1^1 (E - F)) \\ &\sim (a_1^3 (E + F), b_1^1 (E - F)) \\ &\sim (a_1^3 (E + F), a_1^3 (E - F)) \\ &\sim (2a_1^3 E, a_1^3 (E - F)) \\ &\sim (2a_1^3 E, 2a_1^3 (E - F)) \\ &\sim (E, E - F) \end{aligned}$$

Remark. We could have equivalently rotated to H , which would give $(H, E - F)$. It does not matter which one we rotate to. Indeed, we always have $B(A, A) > 0$, hence rotations are all occurring on the hyperboloid of one sheet.

Case 3: $B(B, B) = 0$, B is parabolic

By a Lorentz transformation, B can be reduced to a multiple of E .

$$\begin{aligned} (A, B) &\sim (gAg^{-1}, gBg^{-1}) \\ &= (a_1^1 E + a_2^1 F + a_3^1 H, b_1^1 E) \end{aligned}$$

Again, we find elements of the stabiliser group $Stab(E)$. Such elements are of the form

$$g = \begin{pmatrix} a & b \\ 0 & a \end{pmatrix}, \quad a^2 = 1$$

resembling the null rotation. Applying this to $(a_1^1 E + a_2^1 F + a_3^1 H, b_1^1 E)$,

$$(A, B) \sim ((a_1^1 - b^2 a_2^1 - 2aba_3^1)E + a_2^1 F + (aba_2^1 + a_3^1)H, b_1^1 E)$$

If $a_2^1 = 0$, then

$$\begin{aligned} (A, B) &\sim (a_3^1 H, b_1^1 E) \\ &\sim (a_3^1 H, a_3^1 E) \\ &\sim (H, E) \end{aligned}$$

If $a_2^1 \neq 0$, then choose $b = -a_3^1/aa_2^1$ to eliminate the H component in A , giving

$$\begin{aligned} (A, B) &\sim (a_2^1 F, b_1^1 E) \\ &\sim (a_2^1 F, a_2^1 E) \\ &\sim (F, E) \end{aligned}$$

2.3 Classifying Control Systems on H

We consider the system (3) with $q \in H$ and $A, B \in \mathfrak{h}$. We start by writing A and B as linear combinations of the basis elements

$$A = a_1^0 X + a_2^0 Y + a_3^0 Z, \quad B = b_1^0 X + b_2^0 Y + b_3^0 Z$$

The Heisenberg group is different from the previous two scenarios in that the Killing form is degenerate. Firstly, we may use inner automorphisms. These preserve the X and Y components, only shifting the Z component. Taking

$$g = \begin{pmatrix} 1 & a & c \\ 0 & 1 & b \\ 0 & 0 & 1 \end{pmatrix}$$

and applying this transformation to A and B , we obtain

$$gAg^{-1} = a_1^0 X + a_2^0 Y + (-ba_1^0 + aa_2^0 + a_3^0)Z, \quad gBg^{-1} = b_1^0 X + b_2^0 Y + (-bb_1^0 + ab_2^0 + b_3^0)Z$$

The second type of important transformations are symplectic maps, which, in fact, belong to the outer automorphisms. The symplectic map uses a matrix $T \in Sp(2, \mathbb{R})$ to act by left matrix multiplication on the (X, Y) vector in A and B , and preserves the Z component. Symbolically, we can write the mapping

$$((X, Y), Z) \mapsto (T(X, Y), Z)$$

Note that in the $n = 2$ case, $Sp(2, \mathbb{R}) = SL(2, \mathbb{R})$. So when $T \in Sp(2, \mathbb{R})$, T can be taken as the rotation matrix

$$T = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$$

Doing so allows us to rotate in the X - Y plane.

Consider also the centre of the Lie algebra $\mathfrak{h}$, which is $\mathfrak{z}(\mathfrak{h}) = \{cZ : c \in \mathbb{R}\}$. We try to simplify B first, and to do so, we analyze two cases: either $B \in \mathfrak{z}(\mathfrak{h})$, or $B \notin \mathfrak{z}(\mathfrak{h})$.

Case 1: $B \in \mathfrak{z}(\mathfrak{h})$

Here, $b_3^0 \neq 0$, while $b_1^0 X + b_2^0 Y = 0$, i.e. $b_1^0 = b_2^0 = 0$. Then,

$$\begin{aligned} (A, B) &\sim (gAg^{-1}, gBg^{-1}) \\ &= (a_1^0 X + a_2^0 Y + (-ba_1^0 + aa_2^0 + a_3^0)Z, b_3^0 Z) \\ &\sim (a_1^0 X + a_2^0 Y, b_3^0 Z) \end{aligned}$$

It must be that $(a_1^0, a_2^0) \neq (0, 0)$. Using the rotation matrix T simultaneously on A and B , we can rotate $a_1^0 X + a_2^0 Y$ onto either X or Y , while $b_3^0 Z$ remains unchanged.

$$\begin{aligned}(A, B) &\sim (a_1^1 X, b_3^0 Z) \\ &\sim (a_1^1 X, a_1^1 Z) \\ &\sim (X, Z)\end{aligned}$$

Case 2: $B \notin \mathfrak{z}(\mathfrak{h})$

Here, $(b_1^0, b_2^0) \neq (0, 0)$. b_3^0 may or may not be 0. In the inner automorphism on B ,

$$gBg^{-1} = b_1^0 X + b_2^0 Y + (-bb_1^0 + ab_2^0 + b_3^0)Z$$

Since $(b_1^0, b_2^0) \neq (0, 0)$, we can choose $a, b \in \mathbb{R}$ of the matrix g such that $-bb_1^0 + ab_2^0 + b_3^0 = 0$, so

$$gBg^{-1} = b_1^0 X + b_2^0 Y$$

This can further be rotated on to X by applying the symplectic map to A and B . That is,

$$\begin{aligned}(A, B) &\sim (gAg^{-1}, gBg^{-1}) \\ &= (a_1^0 X + a_2^0 Y + (-ba_1^0 + aa_2^0 + a_3^0)Z, b_1^0 X + b_2^0 Y) \\ &\sim (a_1^1 X + a_2^1 Y + (-ba_1^0 + aa_2^0 + a_3^0)Z, b_1^1 X) \\ &\sim (a_2^1 Y + (-ba_1^0 + aa_2^0 + a_3^0)Z, b_1^1 X)\end{aligned}$$

If $a_2^1 = 0$, then

$$\begin{aligned}(A, B) &= ((-ba_1^0 + aa_2^0 + a_3^0)Z, b_1^1 X) \\ &= ((-ba_1^0 + aa_2^0 + a_3^0)Z, (-ba_1^0 + aa_2^0 + a_3^0)X) \\ &= (Z, X)\end{aligned}$$

If $a_2^1 \neq 0$, then we take another $h \in H$ that is also an element of the stabiliser of X . This is

$$h = \begin{pmatrix} 1 & a' & c' \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \in \text{Stab}(X)$$

We now apply the inner automorphism.

$$\begin{aligned}(A, B) &\sim (a_2^1 Y + (-ba_1^0 + aa_2^0 + a_3^0)Z, b_1^1 X) \\ &\sim (h(a_2^1 Y + (-ba_1^0 + aa_2^0 + a_3^0)Z)h^{-1}, h(b_1^1 X)h^{-1}) \\ &= (a_2^1 Y + (a_2^1 a' + (-ba_1^0 + aa_2^0 + a_3^0))Z, b_1^1 X)\end{aligned}$$

Since $a_2^1 \neq 0$, we can choose the entries of h to be $a' = -(-ba_1^0 + aa_2^0 + a_3^0)/a_2^1$, $c' \in \mathbb{R}$ so that $a_2^1 a' + (-ba_1^0 + aa_2^0 + a_3^0) = 0$. Proceeding with this,

$$\begin{aligned}(A, B) &\sim (a_2^1 Y, b_1^1 X) \\ &\sim (a_2^1 Y, a_2^1 X) \\ &\sim (Y, X)\end{aligned}$$

2.4 Classifying Control Systems on $SE(2)$

We consider the system (3) with $q \in SE(2)$ and $A, B \in \mathfrak{se}(2)$. Begin by writing A and B as linear combinations of the basis elements

$$A = a_1^0 X + a_2^0 Y + a_3^0 Z, \quad B = b_1^0 X + b_2^0 Y + b_3^0 Z$$

For a matrix $g \in SE(2)$, the conjugation group action on the basis elements gives

$$gXg^{-1} = \cos \theta X + \sin \theta Y, \quad gYg^{-1} = -\sin \theta X + \cos \theta Y, \quad gZg^{-1} = yX - xY + Z$$

with $x, y, \theta \in \mathbb{R}$. So on A and B , we have

$$gAg^{-1} = (a_1^0 \cos \theta - a_2^0 \sin \theta + a_3^0 y)X + (a_1^0 \sin \theta + a_2^0 \cos \theta - a_3^0 x)Y + a_3^0 Z$$

$$gBg^{-1} = (b_1^0 \cos \theta - b_2^0 \sin \theta + b_3^0 y)X + (b_1^0 \sin \theta + b_2^0 \cos \theta - b_3^0 x)Y + b_3^0 Z$$

There are two cases to consider: when $B \in \text{Span}(X, Y)$, and when $B \notin \text{Span}(X, Y)$.

Case 1: $B \in \text{Span}(X, Y)$

In this case, $B = b_1^0 X + b_2^0 Y$, with $(b_1^0, b_2^0) \neq 0$ and $b_3^0 = 0$. Since $b_3^0 = 0$, we can disregard x and y , and choose θ so that gBg^{-1} rotates B in the X - Y plane onto X , then simplify.

$$\begin{aligned} (A, B) &\sim (gAg^{-1}, gBg^{-1}) \\ &= ((a_1^0 \cos \theta - a_2^0 \sin \theta + a_3^0 y)X + (a_1^0 \sin \theta + a_2^0 \cos \theta - a_3^0 x)Y + a_3^0 Z, b_1^1 X) \\ &= (a_1^1 X + a_2^1 Y + a_3^0 Z, b_1^1 X) \\ &\sim (a_2^1 Y + a_3^0 Z, b_1^1 X) \end{aligned}$$

If $a_3^0 \neq 0$, then we require elements of $\text{Stab}(X)$, so that conjugation preserves $b_1^1 X$. It is clear that we need $\theta = 0$. Moreover, noting the result of gZg^{-1} , we choose $x = a_3^1/a_3^0$ and $y = 0$ to eliminate the Y component and prevent X from reappearing in A . Thus,

$$\begin{aligned} (A, B) &\sim (a_3^0 Z, b_1^1 X) \\ &\sim (a_3^0 Z, a_3^0 X) \\ &\sim (Z, X) \end{aligned}$$

If $a_3^0 = 0$, then assuming $a_2^1 \neq 0$ (otherwise we obtain the same result as before),

$$\begin{aligned} (A, B) &\sim (a_2^1 Y, b_1^1 X) \\ &\sim (a_2^1 Y, a_2^1 X) \\ &\sim (Y, X) \end{aligned}$$

Case 2: $B \notin \text{Span}(X, Y)$

Here, $b_3^0 \neq 0$. When applying inner automorphisms, we first choose $x = 0$ and $y = 0$ so that we only rotate in the X - Y plane.

$$\begin{aligned} (A, B) &\sim (gAg^{-1}, gBg^{-1}) \\ &= ((a_1^0 \cos \theta - a_2^0 \sin \theta + a_3^0 y)X + (a_1^0 \sin \theta + a_2^0 \cos \theta - a_3^0 x)Y + a_3^0 Z, b_1^1 X + b_3^0 Z) \\ &= (a_1^1 X + a_2^1 Y + a_3^0 Z, b_1^1 X + b_3^0 Z) \end{aligned}$$

If $b_1^1 = 0$, then

$$\begin{aligned} (A, B) &\sim (a_1^1 X + a_2^1 Y + a_3^0 Z, b_3^0 Z) \\ &\sim (a_1^1 X + a_2^1 Y, b_3^0 Z) \end{aligned}$$

Apply a second inner automorphism, choosing $x = 0$, $y = 0$, and $\theta \in \mathbb{R}$ to rotate onto X .

$$\begin{aligned}(A, B) &\sim (a_1^2 X, b_3^0 Z) \\ &\sim (a_1^2 X, a_1^2 Z) \\ &\sim (X, Z)\end{aligned}$$

If $b_1^1 \neq 0$, choose another $g \in SE(2)$ with $\theta = 0$ so that the inner automorphism causes no rotation, only translation. Choosing $x = 0$, $y = -b_1^1/b_3^0$,

$$\begin{aligned}(A, B) &\sim (g(a_1^1 X + a_2^1 Y + a_3^0 Z)g^{-1}, g(b_1^1 X + b_3^0 Z)g^{-1}) \\ &= ((a_1^1 + a_3^0 y)X + (a_2^1 - a_3^0 x)Y + a_3^0 Z, (b_1^1 + b_3^0 y)X - b_3^0 xY + b_3^0 Z) \\ &\sim (a_1^2 X + a_2^1 Y + a_3^0 Z, b_3^0 Z) \\ &\sim (a_1^2 X + a_2^1 Y, b_3^0 Z)\end{aligned}$$

Finally, use one last inner automorphism and set $x = 0$, $y = 0$. $\theta \in \mathbb{R}$ is chosen such that $a_1^2 X + a_2^1 Y$ is rotated on to X .

$$\begin{aligned}(A, B) &\sim (a_1^3 X, b_3^0 Z) \\ &\sim (a_1^3 X, a_1^3 Z) \\ &\sim (X, Z)\end{aligned}$$

2.5 Classifying Control Systems on $SH(2)$

Consider the system (3) with $q \in SH(2)$ and $A, B \in \mathfrak{sh}(2)$. The $SH(2)$ group is similar to $SE(2)$, except that $SH(2)$ is translations and hyperbolic rotations preserving the Lorentzian.

Again, begin by writing A and B as linear combinations of the basis elements

$$A = a_1^0 X + a_2^0 Y + a_3^0 Z, \quad B = b_1^0 X + b_2^0 Y + b_3^0 Z$$

For a matrix $g \in SH(2)$, the conjugation group action on the basis elements gives

$$gXg^{-1} = \cosh \theta X + \sinh \theta Y, \quad gYg^{-1} = \sinh \theta X + \cosh \theta Y, \quad gZg^{-1} = -yX - xY + Z$$

with $x, y, \theta \in \mathbb{R}$. Indeed, these transformations are combinations of hyperbolic rotations determined by θ , and translations determined by x and y . For A and B , we have

$$\begin{aligned}gAg^{-1} &= (a_1^0 \cosh \theta + a_2^0 \sinh \theta - a_3^0 y)X + (a_1^0 \sinh \theta + a_2^0 \cosh \theta - a_3^0 x)Y + a_3^0 Z \\ gBg^{-1} &= (b_1^0 \cosh \theta + b_2^0 \sinh \theta - b_3^0 y)X + (b_1^0 \sinh \theta + b_2^0 \cosh \theta - b_3^0 x)Y + b_3^0 Z\end{aligned}$$

The analysis requires two cases to be considered: $B \in \text{Span}(X, Y)$ and $B \notin \text{Span}(X, Y)$.

Case 1: $B \in \text{Span}(X, Y)$

Here, $b_3^0 = 0$ and $(b_1^0, b_2^0) \neq (0, 0)$. If $b_1^0 \neq 0$ and $b_2^0 = 0$, then

$$\begin{aligned}(A, B) &= (a_1^0 X + a_2^0 Y + a_3^0 Z, b_1^0 X) \\ &\sim (a_2^0 Y + a_3^0 Z, b_1^0 X)\end{aligned}$$

Now, if $a_3^0 = 0$,

$$\begin{aligned}(A, B) &\sim (a_2^0 Y, b_1^0 X) \\ &\sim (a_2^0 Y, a_2^0 X) \\ &\sim (Y, X)\end{aligned}$$

Otherwise, if $a_3^0 \neq 0$, apply a transformation consisting only of a shift in Y . This is done by choosing $\theta = 0$, $y = 0$, and $x = a_2^0/a_3^0$.

$$\begin{aligned}
(A, B) &\sim (g(a_2^0 Y + a_3^0 Z)g^{-1}, g(b_1^0 X)g^{-1}) \\
&= ((a_2^0 - a_3^0 x)Y + a_3^0 Z, b_1^0 X) \\
&= (a_3^0 Z, b_1^0 X) \\
&\sim (a_3^0 Z, a_3^0 X) \\
&\sim (Z, X)
\end{aligned}$$

If $b_1^0 = 0$ and $b_2^0 \neq 0$, then

$$\begin{aligned}
(A, B) &= (a_1^0 X + a_2^0 Y + a_3^0 Z, b_2^0 Y) \\
&\sim (a_1^0 X + a_3^0 Z, b_2^0 Y)
\end{aligned}$$

If $a_3^0 = 0$, then

$$\begin{aligned}
(A, B) &\sim (a_1^0 X, b_2^0 Y) \\
&\sim (a_1^0 X, a_1^0 Y) \\
&\sim (X, Y)
\end{aligned}$$

Otherwise, if $a_3^0 \neq 0$, apply a transformation consisting only of a shift in X . This is done by choosing $\theta = 0$, $x = 0$, and $y = a_1^0/a_3^0$.

$$\begin{aligned}
(A, B) &\sim (g(a_1^0 X + a_3^0 Z)g^{-1}, g(b_2^0 Y)g^{-1}) \\
&= ((a_1^0 - a_3^0 y)X + a_3^0 Z, b_2^0 Y) \\
&= (a_3^0 Z, b_2^0 Y) \\
&\sim (a_3^0 Z, a_3^0 Y) \\
&\sim (Z, Y)
\end{aligned}$$

The most general case is $b_1^0 \neq 0$ and $b_2^0 \neq 0$. We try to use hyperbolic rotations to simplify B . Recall the form of B after a hyperbolic rotation,

$$gBg^{-1} = (b_1^0 \cosh \theta + b_2^0 \sinh \theta)X + (b_1^0 \sinh \theta + b_2^0 \cosh \theta)Y$$

$\tanh$ is a mapping $\mathbb{R} \mapsto (-1, 1)$. To reduce one of the coefficients to 0, we must solve for θ in $\tanh \theta = -b_1^0/b_2^0$ or $-b_2^0/b_1^0$, depending on the ratio of b_1^0 and b_2^0 . There is also no solution when $b_1^0 = b_2^0$. So three different cases need to be analyzed.

If $b_1^0 = b_2^0$, then

$$\begin{aligned}
(A, B) &\sim (gAg^{-1}, gBg^{-1}) \\
&= (a_1^1 X + a_2^1 Y + a_3^0 Z, b_1^0 X + b_1^0 Y)
\end{aligned}$$

Now if $a_3^0 = 0$, then

$$\begin{aligned}
(A, B) &\sim (a_1^1 X + a_2^1 Y, b_1^0 X + b_1^0 Y) \\
&\sim (a_1^2 X, b_1^0(X + Y)) \\
&\sim (a_1^2 X, a_1^2(X + Y)) \\
&\sim (X, X + Y)
\end{aligned}$$

Otherwise, if $a_3^0 \neq 0$, take another $g \in SH(2)$ with $\theta = 0$ so that there is no rotation. Applying the transformation gives

$$(A, B) \sim ((a_1^1 - a_3^0 y)X + (a_2^1 - a_3^0 x)Y + a_3^0 Z, b_1^0(X + Y))$$

Since $a_3^0 \neq 0$, it is possible to choose $x, y \in \mathbb{R}$ such that $a_1^1 - a_3^0 y = 0$ and $a_2^1 - a_3^0 x = 0$. Hence

$$\begin{aligned}(A, B) &\sim (a_3^0 Z, b_1^0(X + Y)) \\ &\sim (a_3^0 Z, a_3^0(X + Y)) \\ &\sim (Z, X + Y)\end{aligned}$$

If $b_1^0 < b_2^0$, then the equation $b_1^0 \cosh \theta + b_2^0 \sinh \theta = 0$ has a solution. Using such a θ , we obtain (X, Y) and (Z, Y) . If $b_2^0 < b_1^0$, then the equation $b_1^0 \sinh \theta + b_2^0 \cosh \theta = 0$ has a solution. Using such a θ , we obtain (Y, X) and (Z, X) . These equivalent systems have already been encountered.

Case 2: $B \notin \text{Span}(X, Y)$

Here, $b_3^0 \neq 0$. Recalling the general form of gAg^{-1} and gBg^{-1} that was initially derived, it is possible to choose $x, y \in \mathbb{R}$ such that the coefficients $b_1^0 \cosh \theta + b_2^0 \sinh \theta - b_3^0 y = 0$ and $b_1^0 \sinh \theta + b_2^0 \cosh \theta - b_3^0 x = 0$, causing the X and Y components in B to vanish. We have

$$\begin{aligned}(A, B) &\sim (gAg^{-1}, gBg^{-1}) \\ &= (a_1^1 X + a_2^1 Y + a_3^0 Z, b_3^0 Z) \\ &\sim (a_1^1 X + a_2^1 Y, b_3^0 Z)\end{aligned}$$

If $a_1^1 \neq 0$ and $a_2^1 = 0$, then

$$\begin{aligned}(A, B) &\sim (a_1^1 X, b_3^0 Z) \\ &\sim (a_1^1 X, a_1^1 Z) \\ &\sim (X, Z)\end{aligned}$$

If $a_1^1 = 0$ and $a_2^1 \neq 0$, then

$$\begin{aligned}(A, B) &\sim (a_2^1 Y, b_3^0 Z) \\ &\sim (a_2^1 Y, a_2^1 Z) \\ &\sim (Y, Z)\end{aligned}$$

If $a_1^1 \neq 0$ and $a_2^1 \neq 0$, then just as before, there are three cases to analyse. If $a_1^1 = a_2^1$, then

$$\begin{aligned}(A, B) &\sim (a_1^1(X + Y), b_3^0 Z) \\ &\sim (a_1^1(X + Y), a_1^1 Z) \\ &\sim (X + Y, Z)\end{aligned}$$

Otherwise, to simplify further, we search for an element $g \in SH(2)$ belonging to the stabiliser group $\text{Stab}(Z)$. It is clear that $x = 0$ and $y = 0$. Applying the inner automorphism,

$$\begin{aligned}(A, B) &\sim (g(a_1^1 X + a_2^1 Y)g^{-1}, g(b_3^0 Z)g^{-1}) \\ &= ((a_1^1 \cosh \theta + a_2^1 \sinh \theta)X + (a_1^1 \sinh \theta + a_2^1 \cosh \theta)Y, b_3^0 Z)\end{aligned}$$

We follow the same reasoning as before and apply rotations to simplify. If $a_1^1 < a_2^1$, then there exists $\theta \in \mathbb{R}$ such that $a_1^1 \cosh \theta + a_2^1 \sinh \theta = 0$. Therefore

$$\begin{aligned}(A, B) &\sim (a_2^2 Y, b_3^0 Z) \\ &\sim (a_2^2 Y, a_2^2 Z) \\ &\sim (Y, Z)\end{aligned}$$

On the other hand, if $a_2^1 < a_1^1$, then there exists $\theta \in \mathbb{R}$ such that $a_1^1 \sinh \theta + a_2^1 \cosh \theta = 0$. Hence

$$\begin{aligned}(A, B) &\sim (a_1^2 X, b_3^0 Z) \\ &\sim (a_1^2 X, a_1^2 Z) \\ &\sim (X, Z)\end{aligned}$$

3 Small-Time Controllability

Assume the equivalent system is (f_0, f_1) , or

$$\dot{q}(t) = (f_0 + uf_1)q(t) \quad (4)$$

with $q \in G$ and $f_0, f_1 \in \mathfrak{g}$. For each of the equivalent systems found, we wish to determine its small-time controllability. Let us first recall some key definitions.

Definition 3.1. Reachable (attainable) sets starting from x_0 :

- the reachable set from x_0 at time $\tau \geq 0$ is

$$\mathcal{A}(\tau, x_0) = \{x_1 \in \mathbb{R}^n : \exists u(\cdot) : [0, \tau] \rightarrow U, x(\tau, x_0, u) = x_1\};$$

- the reachable set from x_0 within time $\tau \geq 0$ is

$$\mathcal{A}(\leq \tau, x_0) = \cup_{t \in [0, \tau]} \mathcal{A}(t, x_0);$$

- the reachable set from x_0 is

$$\mathcal{A}(x_0) = \cup_{t \in [0, \infty)} \mathcal{A}(t, x_0)$$

Definition 3.2. The system (4) is said to be

- controllable if for every $x_0 \in \mathbb{R}^n$, $\mathcal{A}(x_0) = \mathbb{R}^n$;
- small-time controllable if for every $x_0 \in \mathbb{R}^n$ and $\tau > 0$, we have $\mathcal{A}(\leq \tau, x_0) = \mathbb{R}^n$;
- small-time locally controllable at x_0 if x_0 belongs to the interior of $\mathcal{A}(\leq \tau, x_0)$ for every $\tau > 0$.

Remark. We assumed that $x \in \mathbb{R}^n$. However, we could equally apply these definitions for $x \in M$, a smooth connected manifold.

We now state a classic theorem in control theory.

Theorem 3.3 (Rashevsky-Chow). Let M be a smooth manifold and $X^1, \dots, X^m$ be m smooth vector fields on M . Assume that

$$\text{Lie}\{X^1, \dots, X^m\}(x) = T_x M, \quad \forall x \in M$$

Then the control system

$$\dot{x} = \sum_{i=1}^m u_i X^i(x)$$

is locally controllable in any time at every point of M .

Remark. If M is connected, then local controllability implies global controllability. This is useful since all the matrix Lie groups in this paper are connected.

We hope to apply the Rashevsky-Chow theorem to conclude small-time controllability. However, the problem is that the theorem applies only to driftless systems, while the system (4) clearly has a drift term f_0 . Broadly speaking, the strategy to overcome this problem is to employ fast-oscillating controls and study the remaining drift. First, we must check the bracket generating condition. In this case, it means whether

$$\text{span}\{f_0, f_1, [f_0, f_1]\} = \mathfrak{g}$$

If this condition is satisfied, we may proceed. We consider the convexified adjoint orbit.

$$\text{conv}(Ad_{e^{\theta f_1}} f_0) / \mathbb{R} f_1$$

We look for new drift directions in this orbit, characterised by an infinite line through the origin, then repeat the process. In the end, we hope to obtain a system of the form

$$\dot{q}(t) = (u f_0 + v f_1 + w f_2) q(t)$$

This is an extended system that is compatible with the original system (4), i.e. its attainable set is contained in the closure of the attainable set of (4). Notice we no longer have a drift term, rather three fast directions. This puts us in a position to apply the Rashevsky-Chow theorem to conclude small-time controllability.

It is important to realise that the above is a description of a scenario where we successfully characterise the system to be small-time controllable. In practice, there are numerous points at which this process can fail, and we can directly conclude non-small-time controllability, or we must follow another route to prove non-small-time controllability.

Lemma 3.4. If the pair (f_0, f_1) is not bracket generating, then the system 4 is not small-time controllable.

Proof. Suppose (f_0, f_1) is not bracket generating, then the third direction needed to span the space cannot be generated, hence the pair is not small-time controllable. $\square$

Lemma 3.5. Assume there exists a barrier function F such that

1. $F(g(t)) > 0$ for all trajectories $g(t)$, $t > 0$, of the system 4,
2. $F(I) = 0$,
3. there exists a point X close to the identity I where $F(X) < 0$.

Then the system 4 is not small-time controllable.

Proof. Suppose there exists a function F that satisfies these three conditions. Then here exists a point in a neighbourhood of the identity that cannot be reached by any trajectory of the system in positive time. This implies the system is not small-time locally controllable, hence not small-time controllable. $\square$

Proposition 3.6. The small-time controllability of control systems of the form (3) on each 3D unimodular Lie group is stated in Table 1. In particular, the system (F, E) on $SL(2)$ is the unique equivalent system that is small-time controllable.

Proof. The demonstration for this result is given throughout the following sections. $\square$

3.1 Controllability on $SU(2)$

First, $[\sigma_1, \sigma_3] = -\frac{1}{2}\sigma_2$, so

$$\text{span}\{\sigma_1, \sigma_3, [\sigma_1, \sigma_3]\} = \text{span}\{\sigma_1, \sigma_3, -\frac{1}{2}\sigma_2\} = \mathfrak{su}(2)$$

The pair is bracket generating, so we may proceed to the next step. Define the set

$$L = \{\overline{e^{s\sigma_3}} : s \in \mathbb{R}\}$$

Note that in this case, the closure $\overline{e^{s\sigma_3}} = e^{s\sigma_3}$. It is easy to compute

$$Ad_L \sigma_1 = e^{s\sigma_3} \sigma_1 e^{-s\sigma_3} = \{\cos(s)\sigma_1 + \sin(s)\sigma_2 : s \in \mathbb{R}\}$$

We then convexify this set, and quotient out the fast direction σ_3 .

$$\text{conv}(Ad_L \sigma_1) / \mathbb{R} \sigma_3 = \{(\sigma_1, \sigma_2) : \sigma_1^2 + \sigma_2^2 \leq 1\}$$

This set is a disc in the u_1 - u_2 plane bounded by the unit circle. Since the convexified set is bounded, the system (u_1, u_3) is not small-time controllable.

3.2 Controllability on $SL(2)$

3.2.1 (E, H)

$[E, H] = -2E$, so by Lemma 3.4, the system is not small-time controllable.

3.2.2 (F, H)

$[F, H] = 2F$, so by Lemma 3.4, the system is not small-time controllable.

3.2.3 $(E + F, H)$

The convexified set, after we quotient out the “fast” direction H , is the region above one branch of a hyperbola in the E - F plane, given by

$$\text{conv}(Ad_{e^{vH}}(E + F))/\mathbb{R}H = \text{conv}\{e^{2v}E + e^{-2v}F : v \in \mathbb{R}\}$$

Taking the “fast” direction into account again as in the original system, we have

$$\text{conv}(Ad_{e^{vH}}(E + F)) + uH = \text{conv}\{e^{2v}E + e^{-2v}F : v \in \mathbb{R}\} + uH$$

Moving to the limit, we can achieve the positive half of an infinite line in the direction of E , F , or some linear combination of E and F . For E and F , we have

$$[E, H] = -2E \in \text{span}\{E, H\}, \quad [F, H] = 2F \in \text{span}\{F, H\}$$

On the other hand, for a linear combination $\alpha E + \beta F$,

$$[\alpha E + \beta F, H] = -2\alpha E + 2\beta F \notin \text{span}\{\alpha E + \beta F, H\}$$

meaning this choice generates a new direction. With that being said, the fact that we only have a half-line poses a major obstacle in either case.

If we would like to continue to quotient out both H and the new direction, the main problem is that we only have the positive half of the line. We cannot directly employ the inverse of the matrix exponential, for example e^{-uE} , when computing $Ad_{e^{uE+vH}}(E + F)$. Moreover, we also cannot do

$$SL(2)/e^{uE+vH}$$

to check periodicity, because under the restriction $u > 0$, e^{uE+vH} cannot form a subgroup.

We employ the barrier function method instead. Consider the function

$$F = ac + bd$$

Now we transform the matrix representation into a state-space representation. From

$$\dot{g} = (E + F + uH)g$$

we see

$$\begin{cases} \dot{a} &= au + c \\ \dot{b} &= bu + d \\ \dot{c} &= -cu + a \\ \dot{d} &= -du + b \end{cases}$$

Using this, compute the derivative

$$\dot{F}(g(t)) = \dot{a}c + a\dot{c} + \dot{b}d + b\dot{d} = a^2 + b^2 + c^2 + d^2 \geq 0$$

In fact, given the restriction $ad - bc = 1$, it must be that

$$\dot{F}(g(t)) > 0$$

for all trajectories $g(t)$ of the system $(E + F, H)$. We can also check that at the identity,

$$F(I) = 1 \cdot 0 + 0 \cdot 1 = 0$$

Lastly, consider

$$\exp(-\varepsilon F) = \begin{pmatrix} 1 & 0 \\ -\varepsilon & 1 \end{pmatrix},$$

arbitrarily close to the identity. We have

$$F(\exp(-\varepsilon F)) = -\varepsilon < 0, \forall \varepsilon > 0$$

By Lemma 3.5, $(E + F, H)$ is not small-time controllable.

3.2.4 $(E, E - F)$

Convexified set is the region bounded by an ellipse. Hence the system is not small-time controllable.

3.2.5 (H, E)

$[H, E] = 2E$, so by Lemma 3.4, the system is not small-time controllable.

3.2.6 (F, E)

The convexified set, after we quotient out the “fast” direction E , is a straight line not passing through the origin. It is given by

$$\text{conv}(Ad_{e^{vE}}F)/\mathbb{R}E = \{F + vH : v \in \mathbb{R}\}$$

Taking the “fast” direction into account again as in the original system, we have

$$\text{conv}(Ad_{e^{vE}}F) + uE = \{F + uE + vH : u, v \in \mathbb{R}\}$$

Here, we have unlocked a new direction for the control, H . Now, the E and H directions can be made arbitrarily large by taking large u and v . This effectively makes F negligible. We check the bracket

$$[H, E] = 2E \in \text{span}\{H, E\}$$

E and H were unable to generate a new direction. We need to continue to find quotient spaces, which gives us new directions to use. Now we consider a modified system

$$\dot{g} = (F + uE + vH)g$$

By taking the commutator, we know that the (closure of the) reachable set is the same as the original system.

Remark. Ideally, we want $[H, E]$ to not be in the span of H and E , as that would give us a new direction. When it is in the span of H and E , the new system we are considering above is still valid. It is because no new direction was generated that we need to continue to quotient (otherwise, with two fast directions generating the third direction, we would be done).

This new system has two fast directions, E and H , which we quotient from the convexified Ad orbit. This is done by computing

$$\text{conv}(Ad_{e^{uE+vH}}F)/\text{span}\{E, H\} = \{e^{-2v}F : v \in \mathbb{R}\}$$

This one-dimensional set is the positive half of an infinite line in the F direction, not including the origin. It is appropriate to take the closure of the set to include the origin?

Remark. When computing $Ad_{e^{uE+vH}}F$, we need the inverse matrix exponential, e^{-uE-vH} . Mathematically, this is possible for any matrix. However, possible restrictions on the controls mean we need to be more cautious. After the first quotient, the convexified set was an infinite straight line not passing through the origin in the F - H plane. By taking points on the line to the limit (taking large v), we can approximate an infinite line along the H axis passing the origin, that most importantly, goes in both directions. We were able to obtain both directions for free. Thanks to this, we can indeed directly use e^{-uE-vH} in this scenario. This is different from the hyperbola cases, where we only have the positive half of the line.

Let us now check if the solution to the system

$$\dot{g} = (wF + uE + vH)g, \quad w > 0$$

is periodic. If it is, then the periodicity allows this set to be viewed as an infinite line passing through the origin. Suppose we take $v = 0$, $u = -1$, $w = 1$ so that

$$wF + uE + vH = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$$

The solution to this matrix differential equation is

$$\begin{aligned} g(t) &= \exp\left(\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}t\right)g(0) \\ &= [\cos(t)I + \sin(t)(F - E)]g(0) \end{aligned}$$

which is a periodic solution. In fact, the solution will always be periodic given $v = 0$, $u = -w$ with $w > 0$. This suggests that the set we found can be seen as an infinite line through the origin.

Moreover, we can examine

$$SL(2)/e^{uE+vH}$$

where e^{uE+vH} is interpreted as a subgroup of $SL(2)$. The result must be a one-dimensional manifold, which is completely classified as a line $\mathbb{R}$ and a circle S^1 . Using the KAN decomposition, we have

$$SL(2)/e^{uE+vH} \cong S^1$$

This result supports the periodic solution we just found.

3.3 Controllability on H

3.3.1 (X, Z)

$[X, Z] = 0$, so by Lemma 3.4, the system is not small-time controllable.

3.3.2 (Z, X)

$[Z, X] = 0$, so by Lemma 3.4, the system is not small-time controllable.

3.3.3 (Y, X)

The convexified set, after we quotient out the “fast” direction X , is a straight line not passing through the origin. It is given by

$$\text{conv}(Ad_{e^{vX}}Y)/\mathbb{R}X = \{Y + vZ : v \in \mathbb{R}\}$$

Taking the “fast” direction into account again as in the original system, we have

$$\text{conv}(Ad_{e^{vX}}Y) + uX = \{Y + uX + vZ : u, v \in \mathbb{R}\}$$

Now, the X and Z directions can be made arbitrarily large by taking large u and v . This effectively makes Y negligible. We check the bracket

$$[Z, X] = 0 \in \text{span}\{Z, X\}$$

X and Z were unable to generate a new direction. We need to continue to find quotient spaces, which gives us new directions to use. Now we consider a modified system

$$\dot{g} = (Y + uX + vZ)g$$

By taking the commutator, we know that the (closure of the) reachable set is the same as the original system. This new system has two fast directions, X and Z , which we quotient from the convexified Ad orbit. This is done by computing

$$\text{conv}(Ad_{e^{uX+vZ}}Y)/\text{span}\{X, Z\}$$

We compute

$$e^{uX+vZ} = \exp \left(\begin{pmatrix} 0 & u & v \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \right) = \left\{ \begin{pmatrix} 1 & u & v \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} : u, v \in \mathbb{R} \right\}$$

So

$$Ad_{e^{uX+vZ}}Y = \begin{pmatrix} 1 & 0 & u \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix} = \{Y + uZ : u \in \mathbb{R}\}$$

Convexifying this set,

$$\text{conv}(Ad_{e^{uX+vZ}}Y) = \{Y + uZ : u \in \mathbb{R}\}$$

We then quotient out the “fast” X and Z directions, giving

$$\text{conv}(Ad_{e^{uX+vZ}}Y)/\text{span}\{X, Z\} = \{Y\}$$

which is a singleton, a one-dimensional bounded set. This suggests the system is not small-time controllable.

3.4 Controllability on $SE(2)$

3.4.1 (Y, X)

$[Y, X] = 0$, so by Lemma 3.4, the system is not small-time controllable.

3.4.2 (Z, X)

The convexified set, after we quotient out the “fast” direction X , is a straight line not passing through the origin. It is given by

$$\text{conv}(Ad_{e^{vX}}Z)/\mathbb{R}X = \{Z - vY : v \in \mathbb{R}\}$$

Taking the “fast” direction into account again as in the original system, we have

$$\text{conv}(Ad_{e^{vX}}Z) + uX = \{Z + uX - vY : u, v \in \mathbb{R}\}$$

Now, the X and Y directions can be made arbitrarily large by taking large u and v . This effectively makes Z negligible. We check the bracket

$$[Y, X] = 0 \in \text{span}\{Y, X\}$$

X and Y were unable to generate a new direction. We need to continue to find quotient spaces, which gives us new directions to use. Now we consider a modified system

$$\dot{g} = (Z + uX - vY)g$$

By taking the commutator, we know that the (closure of the) reachable set is the same as the original system. This new system has two fast directions, X and Y , which we quotient from the convexified Ad orbit. This is done by computing

$$\text{conv}(Ad_{e^{uX-vY}}Z)/\text{span}\{X, Y\}$$

We compute

$$e^{uX-vY} = \exp \left(\begin{pmatrix} 0 & 0 & u \\ 0 & 0 & -v \\ 0 & 0 & 0 \end{pmatrix} \right) = \left\{ \begin{pmatrix} 1 & 0 & u \\ 0 & 1 & -v \\ 0 & 0 & 1 \end{pmatrix} : u, v \in \mathbb{R} \right\}$$

So

$$Ad_{e^{uX-vY}}Z = \begin{pmatrix} 0 & -1 & -v \\ 1 & 0 & -u \\ 0 & 0 & 0 \end{pmatrix} = \{Z - vX - uY : u, v \in \mathbb{R}\}$$

Convexifying this set gives a plane passing through Z parallel to the X - Y plane. We then quotient out the “fast” X and Y directions, giving

$$\text{conv}(Ad_{e^{uX-vY}}Z)/\text{span}\{X, Y\} = \{Z\}$$

which is a singleton, a one-dimensional bounded set. This suggests the system is not small-time controllable.

3.4.3 (X, Z)

Convexified set is the region bounded by a circle. Hence the system is not small-time controllable.

3.5 Controllability on $SH(2)$

3.5.1 (X, Z)

The convexified set, after we quotient out the “fast” direction Z , is the region to the right of one branch of a hyperbola in the X - Y plane, given by

$$\text{conv}(Ad_{e^{vZ}}X)/\mathbb{R}Z = \text{conv}\{\cosh vX + \sinh vY : v \in \mathbb{R}\}$$

Taking the “fast” direction into account again as in the original system, we have

$$\text{conv}(Ad_{e^{vZ}}X) + uZ = \text{conv}\{\cosh vX + \sinh vY : v \in \mathbb{R}\} + uZ$$

Moving to the limit, we can achieve the positive half of an infinite line in the direction of $X + Y$ or $X - Y$ (along the asymptotes of the hyperbola), or some other linear combination of X and Y that keeps the line within the convexified set. For $X + Y$ and $X - Y$, we have

$$[X + Y, Z] = -Y - X \in \text{span}\{X + Y, Z\}, \quad [X - Y, Z] = -Y + X \in \text{span}\{-X + Y, Z\}.$$

Another obvious choice of direction would be X , with the distinction that

$$[X, Z] = -Y \notin \text{span}\{X, Z\},$$

Or more generally, taking $\alpha > 0$, $\beta > 0$,

$$[\alpha X + \beta Y, Z] = -\alpha Y - \beta X \notin \text{span}\{\alpha X + \beta Y, Z\}$$

unless $\alpha = \beta$, and

$$[\alpha X - \beta Y, Z] = -\alpha Y + \beta X \notin \text{span}\{\alpha X - \beta Y, Z\}$$

unless $\alpha = \beta$. So these choices are able to generate new directions.

Similar to the $(E + F, H)$ case in $SL(2)$, we cannot directly employ the inverse of the matrix exponential when computing the conjugate. We try to apply the barrier function method again. Consider the function

$$F = x \cosh \theta - y \sinh \theta$$

Now, let us transform the matrix representation of the system (X, Z) into the state-space representation. From

$$\dot{g} = (X + uZ)g$$

we have

$$\begin{cases} \dot{x} &= uy + 1 \\ \dot{y} &= ux \\ \dot{\theta} &= u \end{cases}$$

Using the above choice of F ,

$$F(I) = 0$$

at the identity,

$$\dot{F}(g(t)) = \dot{x} \cosh \theta + x \sinh \theta \dot{\theta} - \dot{y} \sinh \theta - y \cosh \theta \dot{\theta} = \cosh \theta \geq 1 > 0$$

for all trajectories $g(t)$ of the system for $t > 0$, and

$$F(\exp(-\varepsilon X)) = F\left(\begin{pmatrix} 1 & 0 & -\varepsilon \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}\right) = -\varepsilon < 0, \quad \forall \varepsilon > 0$$

arbitrarily close to the identity. Hence, by Lemma 3.5, (X, Z) is not small-time controllable.

3.5.2 (Y, Z)

The convexified set, after we quotient out the “fast” direction Z , is the region above one branch of a hyperbola in the X - Y plane, given by

$$\text{conv}(Ad_{e^{vZ}}Y)/\mathbb{R}Z = \text{conv}\{\sinh vX + \cosh vY : v \in \mathbb{R}\}$$

Taking the “fast” direction into account again as in the original system, we have

$$\text{conv}(Ad_{e^{vZ}}Y) + uZ = \text{conv}\{\sinh vX + \cosh vY : v \in \mathbb{R}\} + uZ$$

Moving to the limit, we can achieve the positive half of an infinite line in the direction of $X + Y$ or $-X + Y$ (along the asymptotes of the hyperbola), or some other linear combination of X and Y that keeps the line within the convexified set. For $X + Y$ and $-X + Y$, we have

$$[X + Y, Z] = -Y - X \in \text{span}\{X + Y, Z\}, \quad [-X + Y, Z] = Y - X \in \text{span}\{-X + Y, Z\}.$$

Another obvious choice of direction would be Y , with the distinction that

$$[Y, Z] = -X \notin \text{span}\{Y, Z\},$$

Or more generally, taking $\alpha > 0, \beta > 0$,

$$[\alpha X + \beta Y, Z] = -\alpha Y - \beta X \notin \text{span}\{\alpha X + \beta Y, Z\}$$

unless $\alpha = \beta$, and

$$[-\alpha X + \beta Y, Z] = \alpha Y - \beta X \notin \text{span}\{-\alpha X + \beta Y, Z\}$$

unless $\alpha = \beta$. So these choices are able to generate new directions.

Similar to the $(E + F, H)$ case in $SL(2)$, we cannot directly employ the inverse of the matrix exponential when computing the conjugate. The method we will follow is almost identical to the (X, Z) case.

First, we write the matrix differential equation

$$\dot{g} = (Y + uZ)g$$

in its state-space representation. That is,

$$\begin{cases} \dot{x} &= uy \\ \dot{y} &= ux + 1 \\ \dot{\theta} &= u \end{cases}$$

Consider the function

$$F(g(t)) = y \cosh \theta - x \sinh \theta$$

for which

$$F(I) = 0$$

at the identity,

$$\dot{F}(g(t)) = \dot{y} \cosh \theta + y \sinh \theta \dot{\theta} - \dot{x} \sinh \theta - x \cosh \theta \dot{\theta} = \cosh \theta \geq 1 > 0$$

for all trajectories $g(t)$ of the system for $t > 0$, and

$$F(\exp(-\varepsilon Y)) = F\left(\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & -\varepsilon \\ 0 & 0 & 1 \end{pmatrix}\right) = -\varepsilon < 0, \quad \forall \varepsilon > 0$$

at a point arbitrarily close to the identity. By Lemma 3.5, the system (Y, Z) is not small-time controllable.

3.5.3 $(X + Y, Z)$

$[X + Y, Z] = -X - Y$, so by Lemma 3.4, the system is not small-time controllable.

3.5.4 $(X, X + Y)$

$[X, X + Y] = 0$, so by Lemma 3.4, the system is not small-time controllable.

3.5.5 $(Z, X + Y)$

$[Z, X + Y] = X + Y$, so by Lemma 3.4, the system is not small-time controllable.

3.5.6 (Z, Y)

The convexified set, after we quotient out the “fast” direction Y , is a straight line not passing through the origin. It is given by

$$\text{conv}(Ad_{e^{vY}}Z)/\mathbb{R}Y = \{Z - vX : v \in \mathbb{R}\}$$

Taking the “fast” direction into account again as in the original system, we have

$$\text{conv}(Ad_{e^{vY}}Z) + uY = \{Z + uY - vX : u, v \in \mathbb{R}\}$$

Now, the X and Y directions can be made arbitrarily large by taking large u and v . This effectively makes Z negligible. We check the bracket

$$[X, Y] = 0 \in \text{span}\{X, Y\}$$

X and Y were unable to generate a new direction. We need to continue to find quotient spaces, which gives us new directions to use. Now we consider a modified system

$$\dot{g} = (Z + uY - vX)g$$

By taking the commutator, we know that the (closure of the) reachable set is the same as the original system. This new system has two fast directions, X and Y , which we quotient from the convexified Ad orbit. This is done by computing

$$\text{conv}(Ad_{e^{uY-vX}}Z)/\text{span}\{X, Y\}$$

We compute

$$e^{uY-vX} = \exp \left(\begin{pmatrix} 0 & 0 & -v \\ 0 & 0 & u \\ 0 & 0 & 0 \end{pmatrix} \right) = \left\{ \begin{pmatrix} 1 & 0 & -v \\ 0 & 1 & u \\ 0 & 0 & 1 \end{pmatrix} : u, v \in \mathbb{R} \right\}$$

So

$$Ad_{e^{uY-vX}}Z = \begin{pmatrix} 0 & 1 & -u \\ 1 & 0 & v \\ 0 & 0 & 0 \end{pmatrix} = \{Z - uX + vY : u, v \in \mathbb{R}\}$$

Convexifying this set gives a plane passing through Z parallel to the X - Y plane. We then quotient out the “fast” X and Y directions, giving

$$\text{conv}(Ad_{e^{uY-vX}}Z)/\text{span}\{X, Y\} = \{Z\}$$

which is a singleton, a one-dimensional bounded set. This suggests the system is not small-time controllable.

3.5.7 (X, Y)

$[X, Y] = 0$, so by Lemma 3.4, the system is not small-time controllable.

3.5.8 (Z, X)

The convexified set, after we quotient out the “fast” direction X , is a straight line not passing through the origin. It is given by

$$\text{conv}(Ad_{e^{vX}}Z)/\mathbb{R}X = \{Z - vY : v \in \mathbb{R}\}$$

Taking the “fast” direction into account again as in the original system, we have

$$\text{conv}(Ad_{e^{vX}}Z) + uX = \{Z + uX - vY : u, v \in \mathbb{R}\}$$

Now, the X and Y directions can be made arbitrarily large by taking large u and v . This effectively makes Z negligible. We check the bracket

$$[Y, X] = 0 \in \text{span}\{Y, X\}$$

X and Y were unable to generate a new direction. We need to continue to find quotient spaces, which gives us new directions to use. Now we consider a modified system

$$\dot{g} = (Z + uX - vY)g$$

By taking the commutator, we know that the (closure of the) reachable set is the same as the original system. This new system has two fast directions, X and Y , which we quotient from the convexified Ad orbit. This is done by computing

$$\text{conv}(Ad_{e^{uX-vY}}Z)/\text{span}\{X, Y\}$$

We compute

$$e^{uX-vY} = \exp \left(\begin{pmatrix} 0 & 0 & u \\ 0 & 0 & -v \\ 0 & 0 & 0 \end{pmatrix} \right) = \left\{ \begin{pmatrix} 1 & 0 & u \\ 0 & 1 & -v \\ 0 & 0 & 1 \end{pmatrix} : u, v \in \mathbb{R} \right\}$$

So

$$Ad_{e^{uX-vY}}Z = \begin{pmatrix} 0 & 1 & v \\ 1 & 0 & -u \\ 0 & 0 & 0 \end{pmatrix} = \{Z + vX - uY : u, v \in \mathbb{R}\}$$

Convexifying this set gives a plane passing through Z parallel to the X - Y plane. We then quotient out the “fast” X and Y directions, giving

$$\text{conv}(Ad_{e^{uX-vY}}Z)/\text{span}\{X, Y\} = \{Z\}$$

which is a singleton, a one-dimensional bounded set. This suggests the system is not small-time controllable.

3.5.9 (Y, X)

$[Y, X] = 0$, so by Lemma 3.4, the system is not small-time controllable.

4 Conclusion

Throughout this paper, we first classified all control systems of the form 3 on all 3D unimodular matrix Lie groups, and then determined their small-time controllability. 22 equivalent systems were found in total, of which only one, the system (F, E) on $SL(2)$, was small-time controllable.

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